

Applied Machine Learning and Predictive Modelling 1

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Outline

1 Admin

2 Linear Models

Section 1

Admin

- Nisia Trisconi (weeks 1, 2, 3 and 7)
- Daniel Meister (weeks 4, 6 and 7)
- Dr. Matteo Tanadini (week 5)
- Module responsible: Nisia Trisconi
- Module assistant: Ramón Christen

- 7 lessons (weekly!)
- 08:15 - 11:35
- 45-50 minutes lecture alternated to breaks
- Slides and labs are provided in advance
 - ▶ Labs (in R, but also available in Python) are discussed during the lecture
 - ▶ Slides are there as support (summarise the theory covered in the labs)
- Complementary exercise sheets (solutions available on the weekend)
- Written exam and group work
- 3 ECTS

Office Hours

- Office hours are online on Wednesdays 5-6PM
- Same link as for the lecture :-)
- If nobody shows up in the first 15 minutes, Ramón will leave the zoom meeting
- First office hour: Wednesday, February 25th
- Last office hour: Wednesday, April 8th

Assumed prior knowledge

- Basic statistics
- Linear Models
- Generalised Linear Models (basics)?
- Basic **R**, basic **Python**

- Students familiarise themselves with the main machine learning methods¹ currently used in Data Science
- Focus is on
 - ▶ Application of the statistical learning methods
 - ▶ Correct interpretation of the modelling results

¹ML1 focuses on supervised learning.

- ① Admin + Linear Models
- ② Extending the Linear Model: Non-linearity
- ③ Extending the Linear Model: Generalised Linear Models
- ④ Support Vector Machines
- ⑤ Model Validation
- ⑥ Neural Networks
- ⑦ Big picture + Conclusions

Supporting Material

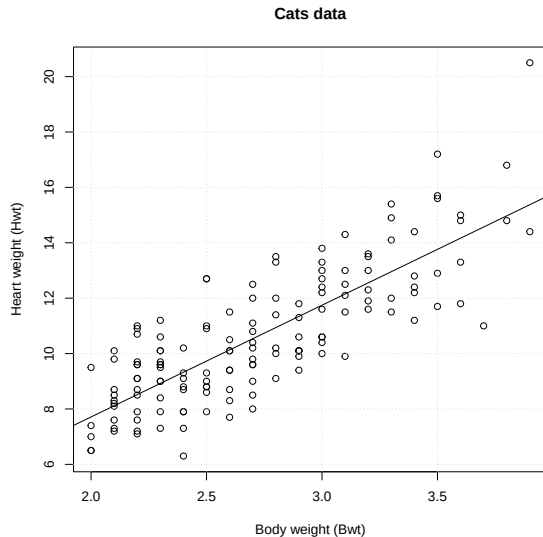
- Book: "An Introduction to Statistical Learning with Applications in R". Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Springer, 2023
- Free book download: <https://www.statlearning.com>
- Book: "An Introduction to Statistical Learning with Applications in Python". Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor, Springer, 2023
- Free book download: <https://www.statlearning.com>
- Video lectures by Hastie & Tibshirani: <https://www.r-bloggers.com/in-depth-introduction-to-machine-learning-in-15-hours-of-expert-videos/>
- NB: **the course does not follow the book very closely. However**, it is a very good idea to complement the course with a book or online material
- The latter comment is especially true in Data Science where very many fields flow together and therefore terminology and views can differ quite dramatically

About me

Section 2

Linear Models

Regression



$$y = \beta_0 + \beta_1 \cdot x_1 + \varepsilon$$

$$Hwt = \beta_0 + \beta_{Bwt} \cdot Bwt + \varepsilon$$

```
lm(Hwt ~ Bwt, data = d.cats)
```

⇒ **See R-code "Cats Lab"**

Regression: Coefficients

- β_0 and β_1 are the regression parameters
- β_0 is the intercept
- β_1 is the slope for the "body weight" predictor
- Regression coefficients are estimated from data
- Regression coefficients are estimated with the "Least Squares" method

Regression: Coefficients

- Estimated regression coefficients are denoted with a hat (e.g. $\hat{\beta}_1$)
- Fitted values (i.e. what the model predicts) are denoted with a hat as well (i.e. $\hat{y}$)
- $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot x_1$
- Residuals are the difference between observed and predicted values
- $\text{res} = y - \hat{y}$

Regression: Coefficients

- If the errors are normally distributed, regression coefficients can be tested with t-tests²
- "Dichotomising" p-values into "significant" / "non-significant" is very bad practice!
- The intercept coefficient is not to be tested. Any linear model must contain an intercept
- **R** uses treatment contrasts to deal with categorical variables

²More about distributional assumptions will come later in this course.

Regression: Coefficient of determination

- The "goodness of fit" can be quantified with R^2
- $R^2 = \text{corr}(y, \hat{y})^2$
- $0 \leq R^2 \leq 1$
- R^2 must not be considered as an "absolute" measure, but rather as case-specific
- R^2 is not used to formally compare models (see F-test later)
- The adjusted R^2 takes into account the complexity of a model (i.e. the number of parameters)

- The p-value quantifies the probability of observing the value of the test statistic, or a more extreme value, under the null hypothesis
- Low p-values are coherent with a rejection of the null hypothesis stating that there is no effect
- For example, in the cats data the p-values for the predictor body weight is extremely small (close to zero) indicating that there is strong evidence that this variable plays a role
- Large p-values (close to 1) do not imply the we can accept the null hypothesis

Testing: Categorical variables

- In **R** categorical variables are called "factors"
- Factors can be tested via F-tests by using the `drop1()` function or by comparing two models via the `anova()` function
- Comparisons among levels of a factor (i.e. "contrasts") can be performed by using the `glht()` function
- Contrasts (usually) imply a p-value correction to avoid the "multiple testing" problem
- Sequential Sums of Squares Anova (i.e. calling the `anova()` function with a single model) should be avoided

Testing: Continuous or discrete variables

- Continuous (and discrete) variables can be tested via F-tests (with `drop1()`) or by t-tests (with `summary()`). The results are fully equivalent
- Sometimes the inferential results for continuous variables are best displayed and communicated with confidence intervals

Testing: Interactions

- If an interaction term is shown to be significant, then all terms involved in this interaction play a relevant role
- An interaction involving two predictors is called "two-fold interaction" (e.g. $\text{age} * \text{species}$)
- An interaction can involve more than two predictors (three-fold, four-fold, ...)
- Parameter interpretation for interaction terms higher than two-fold can be hard

Testing: Model assumptions

- All the testing procedure discussed here rely on assumptions about the errors that will be discussed later in the course
- In particular, errors are assumed to follow a normal distribution
- p-values and confidence intervals, as discussed here, are fully equivalent (duality)³

³For example, a continuous predictor significant at the 5 % level will have a CI that does not contain zero.